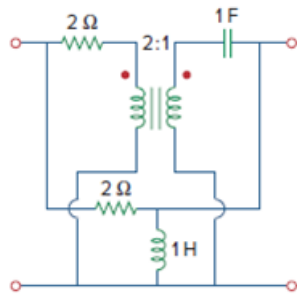


Problem

The circuit in Fig. 19.116 may be regarded as two two-ports connected in parallel. Obtain the y parameters as functions of s .

Figure 19.116



Step-by-step solution

Step 1 of 18

Draw the circuit diagram.

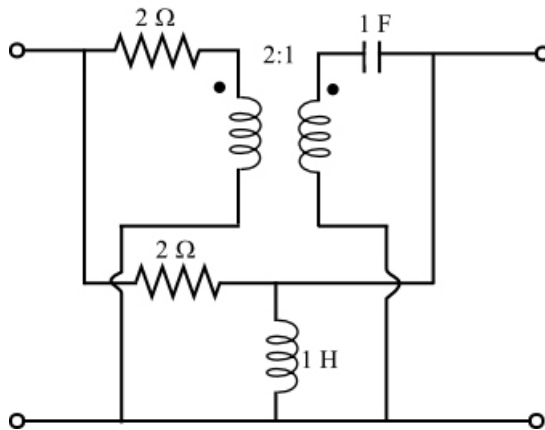


Figure 1

Step 2 of 18

Calculate y -parameters in s domain.

$$y_{11} = \left. \frac{I_1}{V_1} \right|_{V_2=0} \quad y_{12} = \left. \frac{I_1}{V_2} \right|_{V_1=0}$$

$$y_{21} = \left. \frac{I_2}{V_1} \right|_{V_2=0} \quad y_{22} = \left. \frac{I_2}{V_2} \right|_{V_1=0}$$

From Figure 1, divide the network into two parts. Consider the upper network which is drawn as shown below.

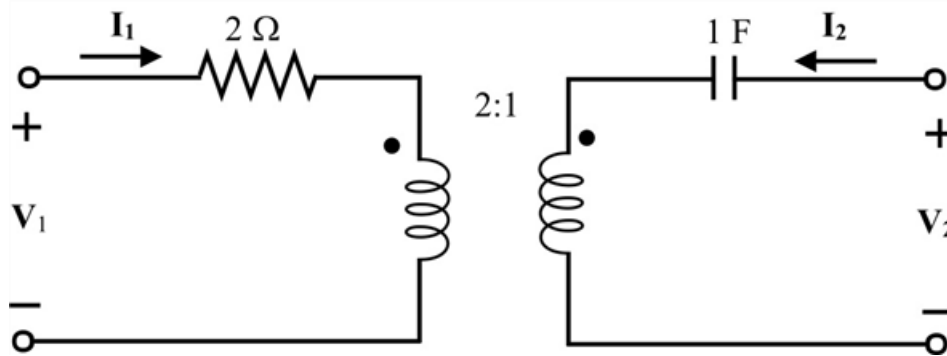


Figure 2

Step 3 of 18

Calculate y_{11} and y_{21} .

Connect a D.C test voltage source of V_1 to the input port and short circuit the output port, that is $V_2 = 0$ V.

Draw Figure 2 in s -domain.

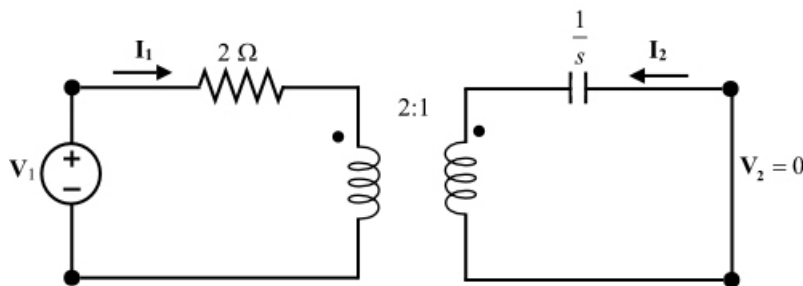


Figure 3

Step 4 of 18

From Figure 3, find the transformation ratio.

$$\begin{aligned}n &= \frac{N_2}{N_1} \\&= \frac{1}{2}\end{aligned}$$

From Figure 3, transform $\frac{1}{s}$ from secondary to primary. Therefore,

$$\begin{aligned}Z_R &= \frac{\frac{1}{s}}{n^2} \\&= \frac{\frac{1}{s}}{\left(\frac{1}{2}\right)^2} \\&= \frac{4}{s}\end{aligned}$$

Write the voltage expression to the primary loop.

$$\begin{aligned}\mathbf{V}_1 &= (2 + Z_R)\mathbf{I}_1 \\&= \left(2 + \frac{4}{s}\right)\mathbf{I}_1 \\&= \left(\frac{2s + 4}{s}\right)\mathbf{I}_1\end{aligned}$$

$$\mathbf{I}_1 = \frac{\mathbf{V}_1}{\left(\frac{2s + 4}{s}\right)}$$

$$\frac{\mathbf{I}_1}{\mathbf{V}_1} = \left(\frac{s}{2s + 4}\right)$$

Therefore, the value of y_{11} is $\left(\frac{s}{2s + 4}\right)$.

Calculate y_{21} .

From transformation ratio, calculate current I_2 .

$$n = \frac{I_1}{I_2}$$

From Figure 3, Consider current I_2 is in negative direction.

$$n = \frac{I_1}{-I_2}$$

Substitute $\frac{1}{2}$ for n and $\frac{V_1}{\left(\frac{2s+4}{s}\right)}$ for I_1 .

$$\frac{1}{2} = \frac{\frac{V_1}{\left(\frac{2s+4}{s}\right)}}{-I_2}$$

$$I_2 = -(2) \frac{V_1}{\left(\frac{2s+4}{s}\right)}$$

$$= -\frac{sV_1}{s+2}$$

$$\frac{I_2}{V_1} = \frac{-s}{s+2}$$

Therefore, the value of y_{21} is $\frac{-s}{s+2}$.

Step 6 of 18

Calculate y_{12} and y_{22} .

Connect a D.C test voltage source of V_2 to the output port and short circuit the input port, that is $V_1 = 0$ V.

Draw the circuit diagram.

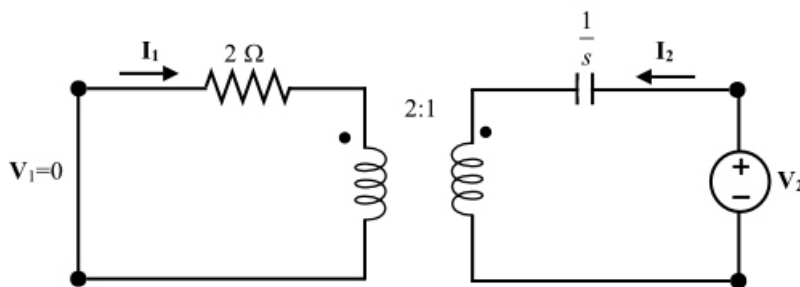


Figure 4

Step 7 of 18

From Figure 4, transform 2Ω from primary to secondary. Therefore,

$$Z_R = n^2 (2 \Omega)$$

$$= \left(\frac{1}{4}\right)(2)$$

$$= \frac{1}{2}$$

Step 8 of 18

Write the voltage expression to the secondary loop.

$$\mathbf{V}_2 = \left(\frac{1}{s} + Z_R \right) \mathbf{I}_2$$

$$\begin{aligned} \mathbf{V}_2 &= \left(\frac{1}{s} + \frac{1}{2} \right) \mathbf{I}_2 \\ &= \left(\frac{s+2}{2s} \right) \mathbf{I}_2 \end{aligned}$$

$$\mathbf{I}_2 = \frac{\mathbf{V}_2}{\left(\frac{s+2}{2s} \right)}$$

$$\frac{\mathbf{I}_2}{\mathbf{V}_2} = \frac{2s}{s+2}$$

Therefore, the value of y_{22} is $\frac{2s}{s+2}$.

Calculate y_{12} .

From transformation ratio, calculate current $\mathbf{I}_1$.

$$n = \frac{\mathbf{I}_1}{\mathbf{I}_2}$$

From Figure 3, Consider current $\mathbf{I}_2$ is in negative direction.

$$n = \frac{\mathbf{I}_1}{-\mathbf{I}_2}$$

$$\mathbf{I}_1 = -n\mathbf{I}_2$$

Step 9 of 18

Calculate $\mathbf{I}_1$.

$$\mathbf{I}_1 = -n\mathbf{I}_2$$

Substitute $\frac{1}{2}$ for n and $\frac{\mathbf{V}_2}{\left(\frac{s+2}{2s} \right)}$ for $\mathbf{I}_2$.

$$\mathbf{I}_1 = -\left(\frac{1}{2} \right) \frac{\mathbf{V}_2}{\left(\frac{s+2}{2s} \right)}$$

$$= \left(\frac{-s}{s+2} \right) \mathbf{V}_2$$

$$\frac{\mathbf{I}_1}{\mathbf{V}_2} = \left(\frac{-s}{s+2} \right)$$

Therefore, the value of y_{12} is $\left(\frac{-s}{s+2} \right)$.

Step 10 of 18

Therefore, the $[y_a]$ parameters are:

$$[y_a] = \begin{bmatrix} \frac{s}{2(s+2)} & \frac{-s}{s+2} \\ \frac{-s}{s+2} & \frac{2s}{s+2} \end{bmatrix}$$

From Figure 1, consider the lower network which is drawn as shown below.

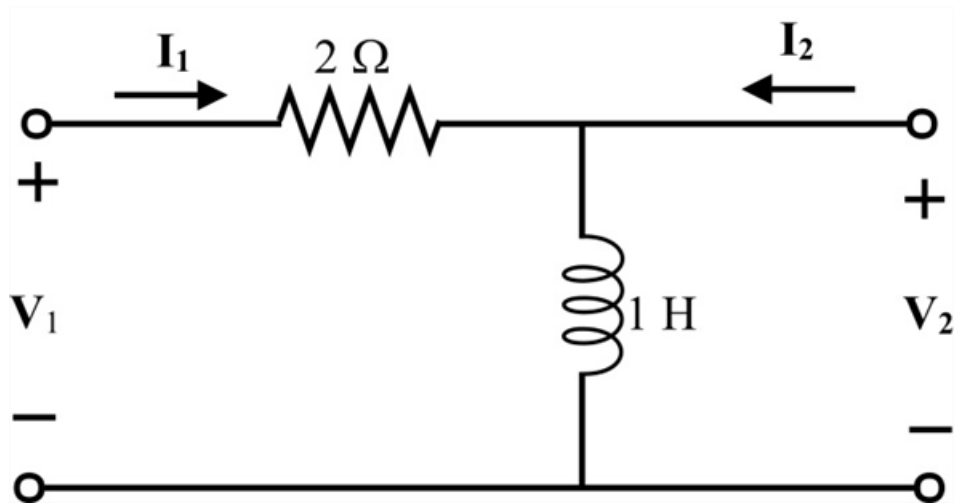


Figure 5

Calculate y_{11} and y_{21} .

Connect a D.C test voltage source of V_1 to the input port and short circuit the output port, that is $V_2 = 0$ V.

Draw Figure 2 in s -domain.

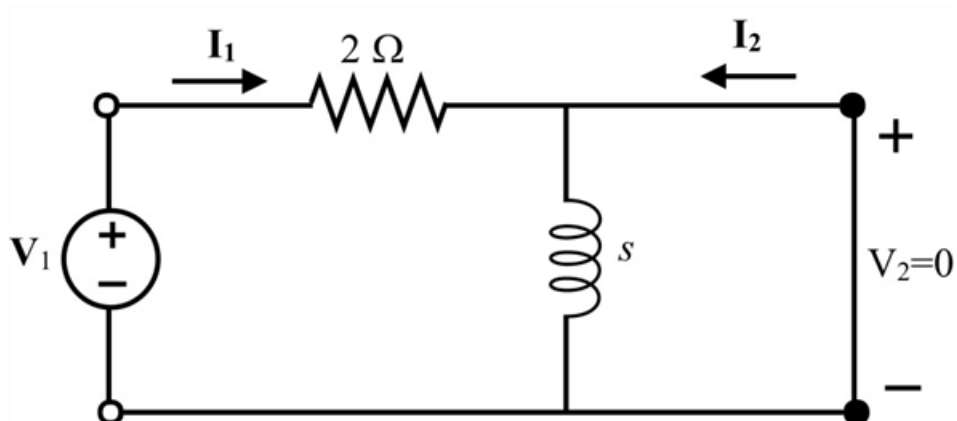


Figure 6

From Figure 6, write the voltage expression to the primary loop.

$$V_1 = (2)(I_1)$$

$$I_1 = \frac{V_1}{2}$$

$$\frac{I_1}{V_1} = \frac{1}{2}$$

Therefore, the value of y_{11} is $\frac{1}{2}$.

Calculate y_{21} .

From Figure 6, calculate the current I_2 .

$$I_2 = -I_1$$

$$= \frac{-V_1}{2}$$

$$\frac{I_2}{V_1} = \frac{-1}{2}$$

Therefore, the value of y_{21} is $\frac{-1}{2}$.

Step 14 of 18

Calculate y_{12} and y_{22} .

Connect a D.C test voltage source of V_2 to the output port and short circuit the input port, that is $V_1 = 0$ V.

Draw the circuit diagram.

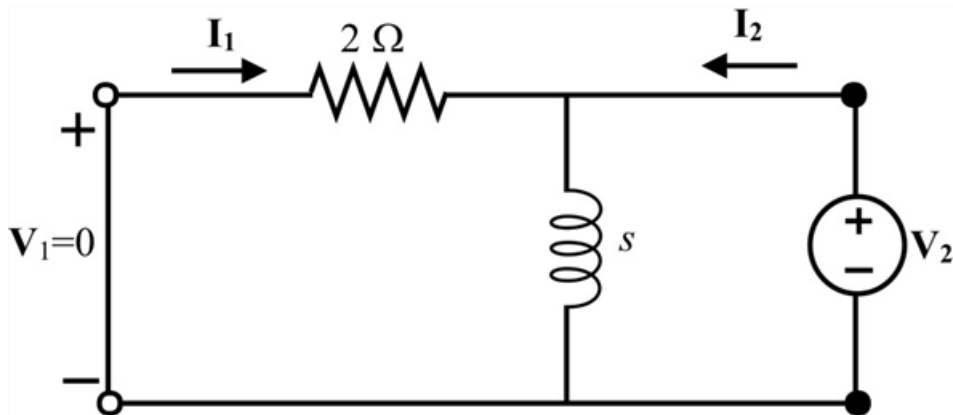


Figure 7

Step 15 of 18

Write the voltage expression to the secondary loop.

$$V_2 = (s \parallel 2) I_2$$

$$= \left(\frac{2s}{s+2} \right) I_2$$

$$I_2 = \left(\frac{s+2}{2s} \right) V_2$$

$$\frac{I_2}{V_2} = \frac{s+2}{2s}$$

Therefore, the value of y_{22} is $\frac{s+2}{2s}$.

From Figure 7, calculate the current I_1 .

$$\begin{aligned}
 I_1 &= -I_2 \left(\frac{s}{s+2} \right) \\
 &= I_2 \left(\frac{-s}{s+2} \right) \\
 &= \left(\frac{-s}{s+2} \right) \left(\frac{s+2}{2s} \right) V_2 \\
 &= \frac{-V_2}{2} \\
 \frac{I_1}{V_2} &= \frac{-1}{2}
 \end{aligned}$$

Therefore, the value of y_{12} is $\frac{-1}{2}$.

Step 17 of 18

Therefore, the $[y_b]$ parameters are:

$$[y_b] = \begin{bmatrix} \frac{1}{2} & \frac{-1}{2} \\ \frac{-1}{2} & \frac{(s+2)}{2s} \end{bmatrix}$$

Step 18 of 18

Calculate the y -parameter of the entire network shown in Figure 1.

$$\begin{aligned}
 [y] &= [y_a] + [y_b] \\
 &= \begin{bmatrix} \frac{s}{2(s+2)} & \frac{-s}{s+2} \\ \frac{-s}{s+2} & \frac{2s}{s+2} \end{bmatrix} + \begin{bmatrix} \frac{1}{2} & \frac{-1}{2} \\ \frac{-1}{2} & \frac{(s+2)}{2s} \end{bmatrix} \\
 &= \begin{bmatrix} \frac{s+1}{s+2} & \frac{-(3s+2)}{2(s+2)} \\ \frac{-(3s+2)}{2(s+2)} & \frac{5s^2+4s+4}{2s(s+2)} \end{bmatrix}
 \end{aligned}$$

Therefore, the y -parameter as function of s is

$$\begin{bmatrix} \frac{s+1}{s+2} & \frac{-(3s+2)}{2(s+2)} \\ \frac{-(3s+2)}{2(s+2)} & \frac{5s^2+4s+4}{2s(s+2)} \end{bmatrix}$$